

AUTOMATION AND ROBOTIC TUTORIAL 1

1. Look at the following control functions. Are they linear/non-linear, time invariant/time variant?

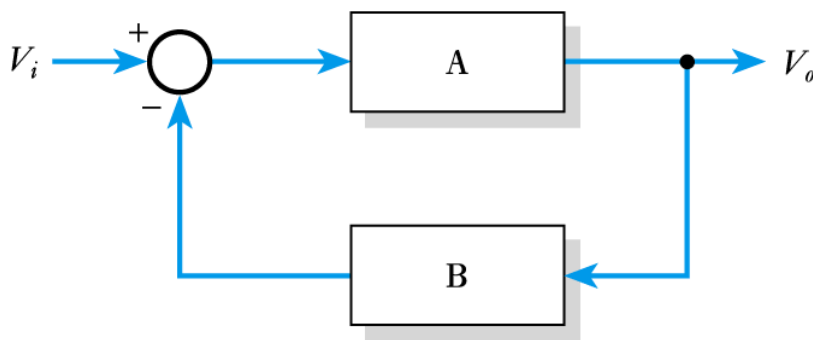
$$y(t) + 2u^2(t) + 7u(t) + 4t$$

$$y(t) + 4u(t) + 7c + 4$$

$$y(t) + 2tu^3(t) + 3u(t)$$

2. Feedback Calculation

Using the standard feedback arrangement from the lecture:



Calculate the required values to produce a stable voltage gain of 100 using a high-gain active amplifier with a gain of 100,000.

Determine the effect on the overall gain of the circuit if the gain of the active amplifier varies from 100,000 to 200,000.